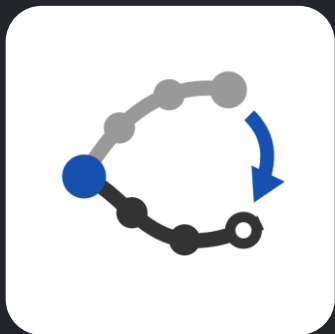




Map curve

Carry a span from one chain onto another by arc length from a reference point: s mm along the first from its zero lands s mm along the second from its zero.



What it does

The same stations are needed on different curves of a ski -- the RSL, the side profile, the core bottom -- measured along each curve from the same place (the FCP). Trim and split work by position, not by distance along the curve. Map curve measures along both chains from a zero point.

Use it when

- A span defined on one curve is needed on another
- Distances must follow the curve, not X
- Arcs must stay arcs (Trim mode never refits)

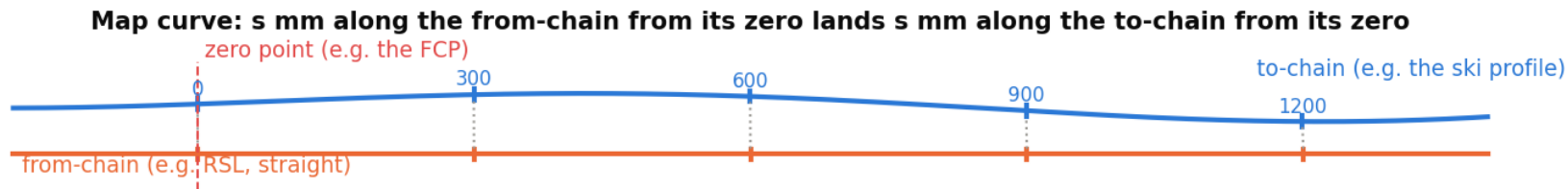
Map curve (Trim to-edges): SKI_PROFILE (grey) trimmed to the RSL span measured along it from the FCP (blue)
RSL Wire (from-edges: sets the span, orange)
result: the same arc length along SKI_PROFILE (blue)



SKI_PROFILE trimmed to the RSL span, measured along it from the FCP.



Arc length from a zero point



Same distance along each

The from-chain's span [start, end], measured from its zero, is laid onto the to-chain from its zero.

World X or closest

World X: the zero lands where each chain crosses the point's X. Closest point: the nearest point on each chain.

Direction check

If the chains run opposite ways at the zero, the feature stops and asks for Flip to-chain.



The dialog

Map curve: SKI_PROFILE... ☒ ☐

Deform from-edges onto to-edges

1 **Trim to-edges to the from span**

Deform and merge to one curve

2 From edges
RSL_Wire

3 To edges
SKI_PROFILE

4 **Shared** Separate

5 Zero point
Vertex of Derived 1

6 **World X** Closest point

8 Name SKI_PROFILE_RSL

9 > Debug

Final

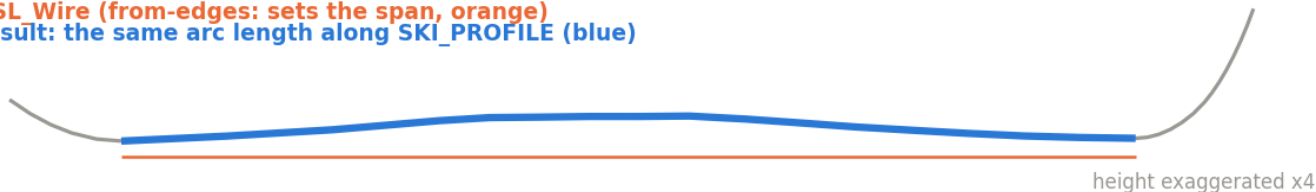
#	Parameter	What it means
1	Mode	Deform from-edges onto to-edges / TRIM to-edges to the from span / Deform and merge to one curve.
2	From edges	The chain that sets the span.
3	To edges	The chain the span lands on (never modified).
4	Reference point	SHARED: one zero point for both. SEPARATE: a zero for each chain.
5	Zero point	A mate connector or vertex (e.g. the FCP).
6	Projection	WORLD X (default) or CLOSEST POINT: how the zero lands on each chain.
7	Flip to-chain	The arrow: when the two chains run opposite ways at the zero.
8	Name	Names the result.
9	Debug	Chain ends (green start, red end) and a summary.



Example: trim to the RSL span

Map curve (Trim to-edges): SKI_PROFILE (grey) trimmed to the RSL span measured along it from the FCP (blue)

RSL Wire (from-edges: sets the span, orange)
result: the same arc length along SKI_PROFILE (blue)



SKI_PROFILE (grey) trimmed to the span of RSL_Wire (orange), zero at the FCP vertex -- result in blue

Trim mode copies the to-edges and cuts them exactly at the two arc lengths; nothing is refitted, so the profile's arcs stay arcs.



Outputs and messages

Published for Extract variables

Key	What it holds
startVertex / endVertex	The ends of the result (start = the tail end: arc length grows with world X).
startEdge / endEdge	The edge at each end.
spanStart / spanEnd	The from-chain's span, measured from its zero.
toStart / toEnd	Where that span lands on the to-chain.
stationCount	Samples used (Deform modes).

Messages

	Message / what to do
Error	chains point in opposite directions ... Tick Flip to-chain
Error	from-edges run X mm past the start / end of the to-edges The to-chain is too short for the span.
Info	do not span the zero point's X; closest point used World X fell back to Closest point.
Info	Merged result is a spline; arc radius not preserved Deform and merge mode.



Tips and limits



Use Trim for exact results

It cuts the original edges; lines and arcs keep their type.



Deform modes are phase 1

They give the to-chain over the from span, resampled and refitted; carrying geometry that lies off the from-chain is not built yet.



Where it lives

Document	Curve_tools (2143812a99089658c704f0bc)
Tab	map_curve (uses curve_core)
Tests	Tests studio (curve_tools_tests: trim to-edges, ends)
Examples	Test_1 studio
Explainer	docs/explainers/curve_tools/curve_tools_explained.md, section 2.2
Related	Clean wire, Map curve, Merge curve, Evaluate profiles, Fillet wire; Extract variables